

Markscheme

Mathematics: analysis and approaches
Practice question bank

98. (a)

It can be proven by induction that $7^n - 1$ is divisible by 6 for every positive integer n .

Using this result, find the value of $\frac{7^5 - 1}{6}$.

$$7^5 = 16\,807 \quad (M1)$$

$$7^5 - 1 = 16\,806 \quad A1$$

$$\frac{16\,806}{6} \quad (M1)$$

$$= 2801 \quad A1$$

[5 marks]